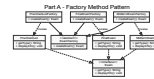


Online Examination Subsystem

UML Class Diagrams for Creational Design Patterns Assignment

Part A: Factory Method Pattern

Exam creation is delegated to concrete factory subclasses. The client depends on ExamFactory and Exam, not direct product construction.



Part B: Abstract Factory Pattern

Each concrete question factory creates a related family: question, renderer, and evaluator. The factories also receive a QuestionSource as required by Part D.



Part C: Builder Pattern

ExamBuilder constructs an immutable ExamConfiguration object through chained setup methods and a final build() call.



Part D: Question Sourcing Integration

QuestionSource abstracts where questions come from. Bank, faker, and manual sources can be swapped without changing the question factory interface.



Note: The current implementation uses `getQuestion(String type)`, which keeps one source object capable of serving multiple question families.